# QUBO Formulation: Mars Satellite Observation Scheduling Problem

## Mars Satellite Observation Scheduling Problem

**Problem Statement.** Given a set of $N$ candidate observations for a Mars-orbiting instrument, each observation $i \in \{0, \ldots, N-1\}$ is associated with a science priority score $p_i \geq 0$, a resource usage vector $\mathbf{u}_i \in \mathbb{R}^K_{\geq 0}$ across $K$ distinct resource types, and a set of temporal or instrumental conflicts $C \subseteq \{(i,j) : i \neq j\}$. Each resource type $k \in \{0, \ldots, K-1\}$ has a total available capacity $R_k > 0$. The goal is to select a subset of observations that maximizes the total science priority score while ensuring that no two selected observations conflict and that the total resource consumption for each type $k$ does not exceed $R_k$.

**Decision Variables.** For each candidate observation $i \in \{0, \ldots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that observation $i$ is selected for execution, and $x_i = 0$ indicates it is not.

**QUBO Derivation.** Step 1 — The original optimization problem is a maximization. To cast it as a minimization problem suitable for quantum annealing or QUBO solvers, we negate the objective function. The unconstrained minimization objective is:

$$\min_{\mathbf{x} \in \{0,1\}^N} \left( -\sum_{i=0}^{N-1} p_i x_i \right). \tag{1}$$

Step 2 — The problem constraints are expressed in general algebraic form. Conflict avoidance requires that for every conflicting pair $(i,j) \in C$, at most one observation is selected:

$$x_i + x_j \leq 1, \quad \forall (i,j) \in C. \tag{2}$$

Since $x_i, x_j \in \{0,1\}$, this is equivalent to the quadratic equality constraint $x_i x_j = 0$. Resource capacity limits require that for each resource type $k \in \{0, \ldots, K-1\}$:

$$\sum_{i=0}^{N-1} u_{i,k} x_i \leq R_k. \tag{3}$$

Step 3 — We convert each constraint into a quadratic penalty term. For conflicts, we add a penalty $A > 0$ for every violated pair:

$$H_{\text{conflict}}(\mathbf{x}) = A \sum_{(i,j)\in C} x_i x_j. \tag{4}$$

For resource constraints, we use a squared penalty $B > 0$. Penalizing the squared deviation from the limit enforces the constraint when $B$ is sufficiently large:

$$H_{\text{resource}}(\mathbf{x}) = B \sum_{k=0}^{K-1} \left( \sum_{i=0}^{N-1} u_{i,k} x_i - R_k \right)^2. \tag{5}$$

Expanding the squared term using the idempotent property $x_i^2 = x_i$ for binary variables:

$$\left( \sum_{i=0}^{N-1} u_{i,k} x_i - R_k \right)^2 = \sum_{i=0}^{N-1} u_{i,k}^2 x_i - 2R_k \sum_{i=0}^{N-1} u_{i,k} x_i + \sum_{i=0}^{N-1} \sum_{j\neq i} u_{i,k} u_{j,k} x_i x_j + R_k^2. \tag{6}$$

Multiplying by $B$ and summing over $k$, the resource penalty contribution becomes:

$$H_{\text{resource}}(\mathbf{x}) = \sum_{i=0}^{N-1} \left[ B \sum_{k=0}^{K-1} (u_{i,k}^2 - 2R_k u_{i,k}) \right] x_i + \sum_{0\leq i<j\leq N-1} \left[ 2B \sum_{k=0}^{K-1} u_{i,k} u_{j,k} \right] x_i x_j + B \sum_{k=0}^{K-1} R_k^2. \tag{7}$$

Step 4 — Combining the negated objective, conflict penalty, and resource penalty yields the total QUBO Hamiltonian:

$$H(\mathbf{x}) = -\sum_{i=0}^{N-1} p_i x_i + A \sum_{(i,j)\in C} x_i x_j + H_{\text{resource}}(\mathbf{x}). \tag{8}$$

Substituting the expanded resource penalty and grouping terms by monomial:

$$H(\mathbf{x}) = \sum_{i=0}^{N-1} \left[ -p_i + B \sum_{k=0}^{K-1} (u_{i,k}^2 - 2R_k u_{i,k}) \right] x_i + \sum_{0\leq i<j\leq N-1} \left[ A \cdot \mathbb{I}((i,j) \in C) + 2B \sum_{k=0}^{K-1} u_{i,k} u_{j,k} \right] x_i x_j + B \sum_{k=0}^{K-1} \tag{9}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

Step 5 — Reading off the QUBO matrix entries $Q_{i,j}$ for $i \leq j$ and the constant offset $c$ in the standard polynomial representation $H(\mathbf{x}) = \sum_{i\leq j} Q_{i,j} x_i x_j + c$:

$$Q_{i,i} = -p_i + B \sum_{k=0}^{K-1} (u_{i,k}^2 - 2R_k u_{i,k}), \quad \forall i \in \{0, \ldots, N-1\}, \tag{10}$$

$$Q_{i,j} = A \cdot \mathbb{I}((i,j) \in C) + 2B \sum_{k=0}^{K-1} u_{i,k} u_{j,k}, \quad \forall 0 \leq i < j \leq N-1, \tag{11}$$

$$c = B \sum_{k=0}^{K-1} R_k^2. \tag{12}$$

**Penalty Weight Justification.** The penalty weights $A$ and $B$ must be chosen sufficiently large to ensure that any violation of the constraints increases the energy more than the maximum possible gain in the objective function. Specifically, let $P_{\max} = \max_i p_i$ and $P_{\text{total}} = \sum_i p_i$. To guarantee that conflict avoidance is strictly enforced, we require $A > P_{\max}$. This ensures that the energy cost of selecting two conflicting observations exceeds the priority score of any single observation. For resource constraints, let $\Delta_{\min}$ be the minimum positive violation amount in resource units. We require $B > P_{\text{total}}/\Delta_{\min}^2$. This guarantees that the quadratic penalty for any resource violation strictly dominates the total possible science priority gain, thereby forcing the global minimum to satisfy all constraints.

**Correctness Argument.** Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal observation schedule because feasible solutions incur zero penalty, reducing $H(\mathbf{x})$ to $-\sum_i p_i x_i + c$. Thus, among feasible solutions, minimizing $H$ is equivalent to maximizing the total priority score. Infeasible solutions incur strictly positive penalty terms. Given the lower bounds on $A$ and $B$, the energy of any infeasible solution exceeds the energy of the best feasible solution. Consequently, the global minimum of $H(\mathbf{x})$ must correspond to a feasible assignment that maximizes the original objective, guaranteeing correctness.

**Illustrative Example.** Consider a small instance with $N = 3$ candidate observations and $K = 1$ resource type. Let the priority scores be $p_0 = 10$, $p_1 = 8$, $p_2 = 5$. The resource usage vector is $\mathbf{u} = [2, 3, 1]^\top$ and the capacity limit is $R_0 = 4$. All pairs conflict, so $C = \{(0,1), (0,2), (1,2)\}$. We set penalty weights $A = 50$ and $B = 300$. Using the general formulas, the concrete objective is $\min -10x_0 - 8x_1 - 5x_2$. The concrete conflict penalty is $50(x_0x_1 + x_0x_2 + x_1x_2)$. The concrete resource penalty is $300(2x_0 + 3x_1 + x_2 - 4)^2$. The resulting QUBO Hamiltonian in expanded form is:

$$H(x_0, x_1, x_2) = -3710x_0 - 4508x_1 - 2105x_2 + 1815x_0x_1 + 615x_0x_2 + 915x_1x_2 + 4800. \quad (13)$$

Evaluating all $2^3 = 8$ binary assignments, the feasible solutions (satisfying $x_i + x_j \leq 1$ and $2x_0 + 3x_1 + x_2 \leq 4$) are $\mathbf{x} = (1,0,0)$, $(0,1,0)$, $(0,0,1)$, and $(0,0,0)$. Their energies are $H(1,0,0) = 1090$, $H(0,1,0) = 292$, $H(0,0,1) = 2695$, and $H(0,0,0) = 4800$. The minimum energy is 292 at $\mathbf{x}^* = (0,1,0)$, corresponding to selecting observation 1 with a total priority score of 8. All infeasible assignments yield energies strictly greater than 292, confirming the formulation correctly identifies the optimal schedule.